Ad Click Prediction ? Executive Summary

Business Objective

Predict rare ad clicks so ad spend shifts toward impressions with higher intent while reducing irrelevant ad exposure.

Data Used

Training rows: 463,291; test rows: 128,858; observed click rate: 6.76%.

Final Model

Selected Random Forest - balanced. Threshold: 0.458; ROC-AUC: 0.574; F1: 0.142; recall: 0.560; precision: 0.081.

Key Signal

Smoothed CTR aggregates and interaction features convert historical user, product, campaign, and webpage behavior into intent indicators.

Deployment Choice

Use class weighting plus threshold tuning for serving. Keep SMOTE as an offline diagnostic because it expands training data size.

Methodology and Reproducible Pipeline

? Loaded train/test CSV files and parsed DateTime into hour, day-of-week, weekend, night, and business-hour signals.

? Handled missing values with categorical and numeric imputers inside sklearn pipelines.

? Engineered personalized user-product, campaign-product, webpage-product, and smoothed CTR aggregate features.

? Used temporal validation: older dates for development and latest date for holdout, matching future ad serving.

? Compared balanced Logistic Regression, balanced Random Forest, weighted XGBoost, and lightweight SMOTE.

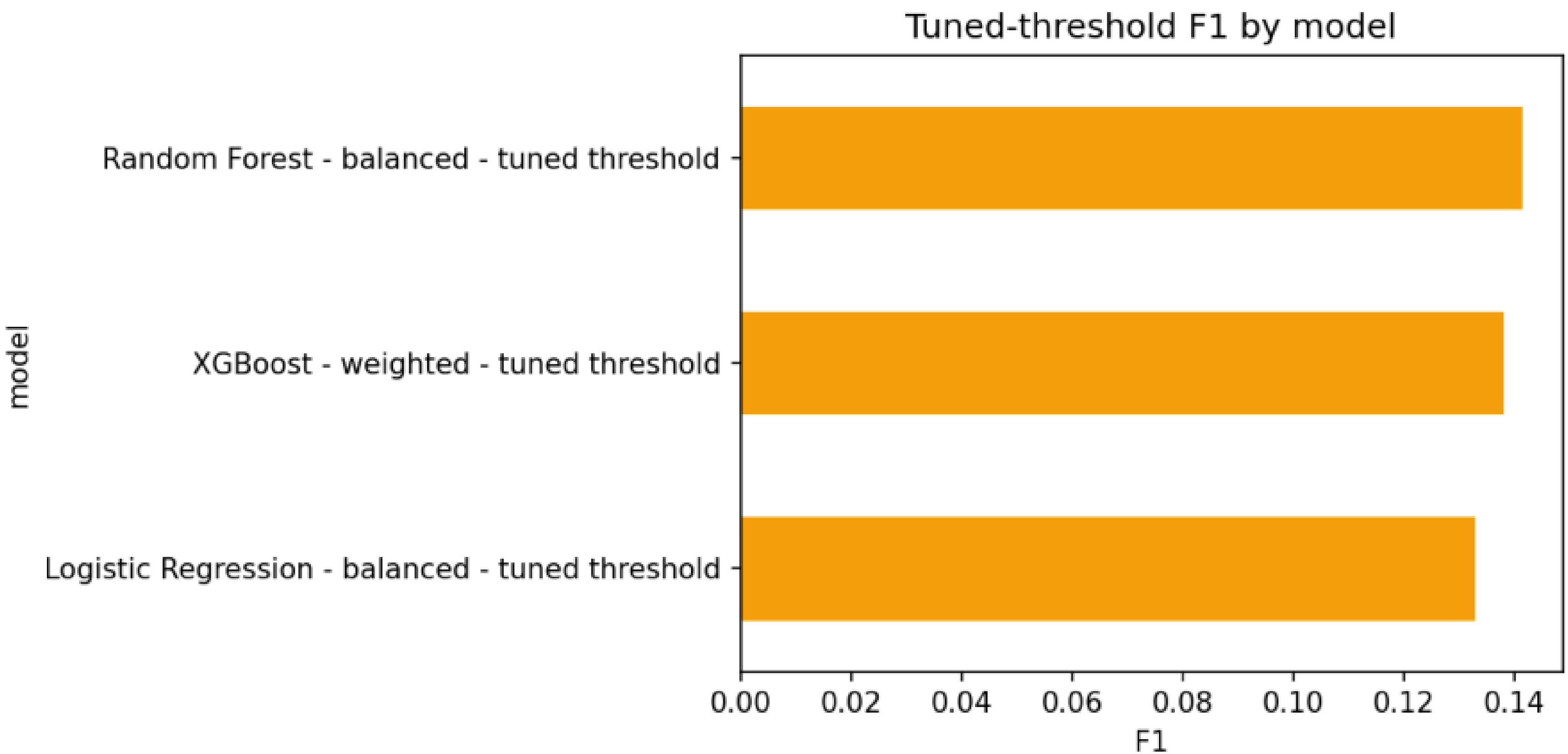
? Generated ad_click_prediction_test_predictions.csv with session-level click probabilities and labels.

Model Comparison on Temporal Holdout

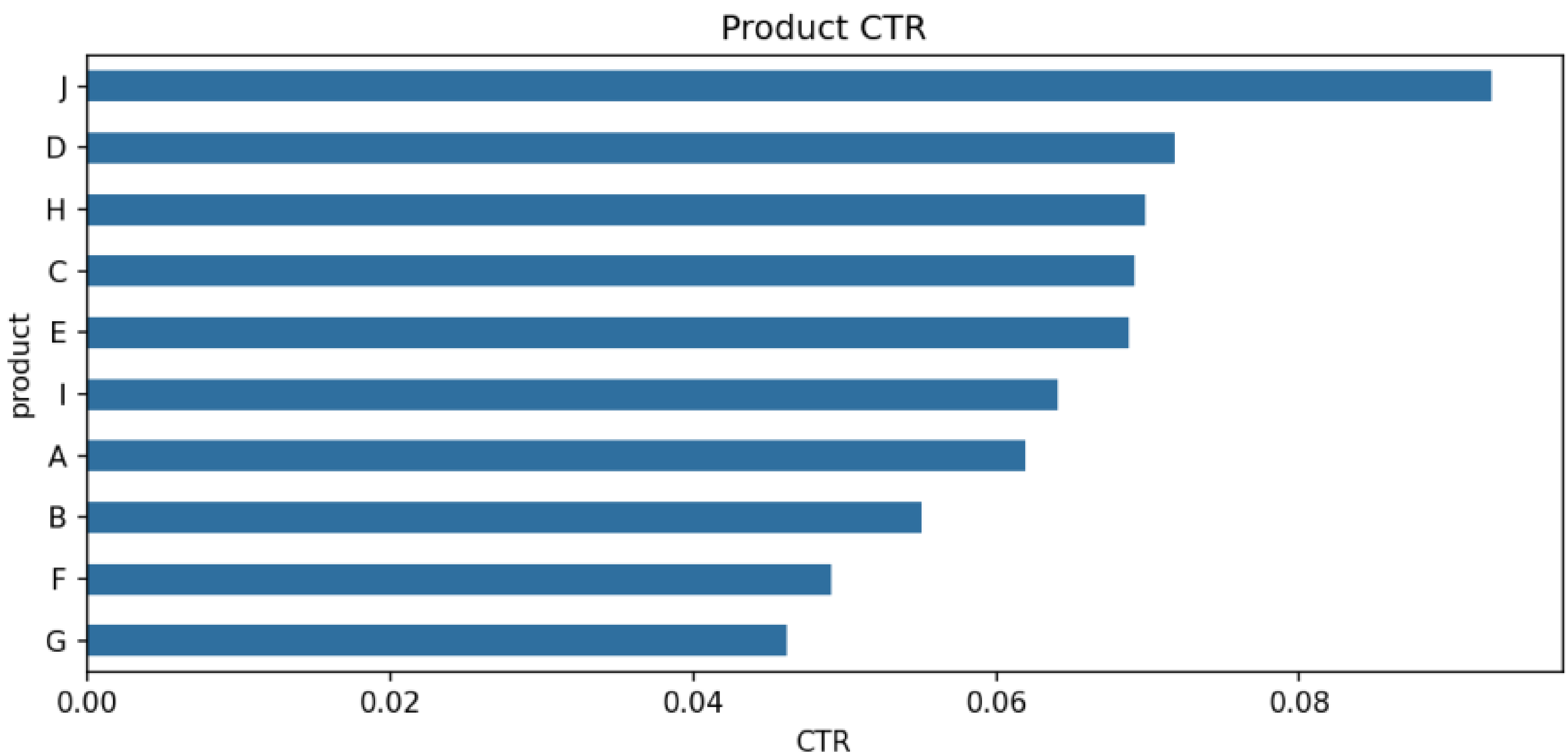
model	threshold	roc_auc	pr_auc	f1	precision	recall	tn	fp	fn	tp
Random Forest - balanced - tu	0.4584	0.5738	0.0738	0.1417	0.0811	0.5603	39122	27943	1935	2466
XGBoost - weighted - tune	0.4762	0.5664	0.0760	0.1381	0.0805	0.4876	42537	24528	2255	2146
XGBoost - weight	0.5000	0.5664	0.0760	0.1352	0.0829	0.3672	49179	17886	2785	1616
Random Forest - bala	0.5000	0.5738	0.0738	0.1343	0.0785	0.4660	42974	24091	2350	2051
ogistic Regression - balanced -	0.0716	0.5764	0.0750	0.1329	0.0740	0.6544	31016	36049	1521	2880
Logistic Regression - b	0.5000	0.5764	0.0750	0.0779	0.0720	0.0848	62260	4805	4028	373

Ranking favors F1/recall because false negatives represent missed monetization opportunities.

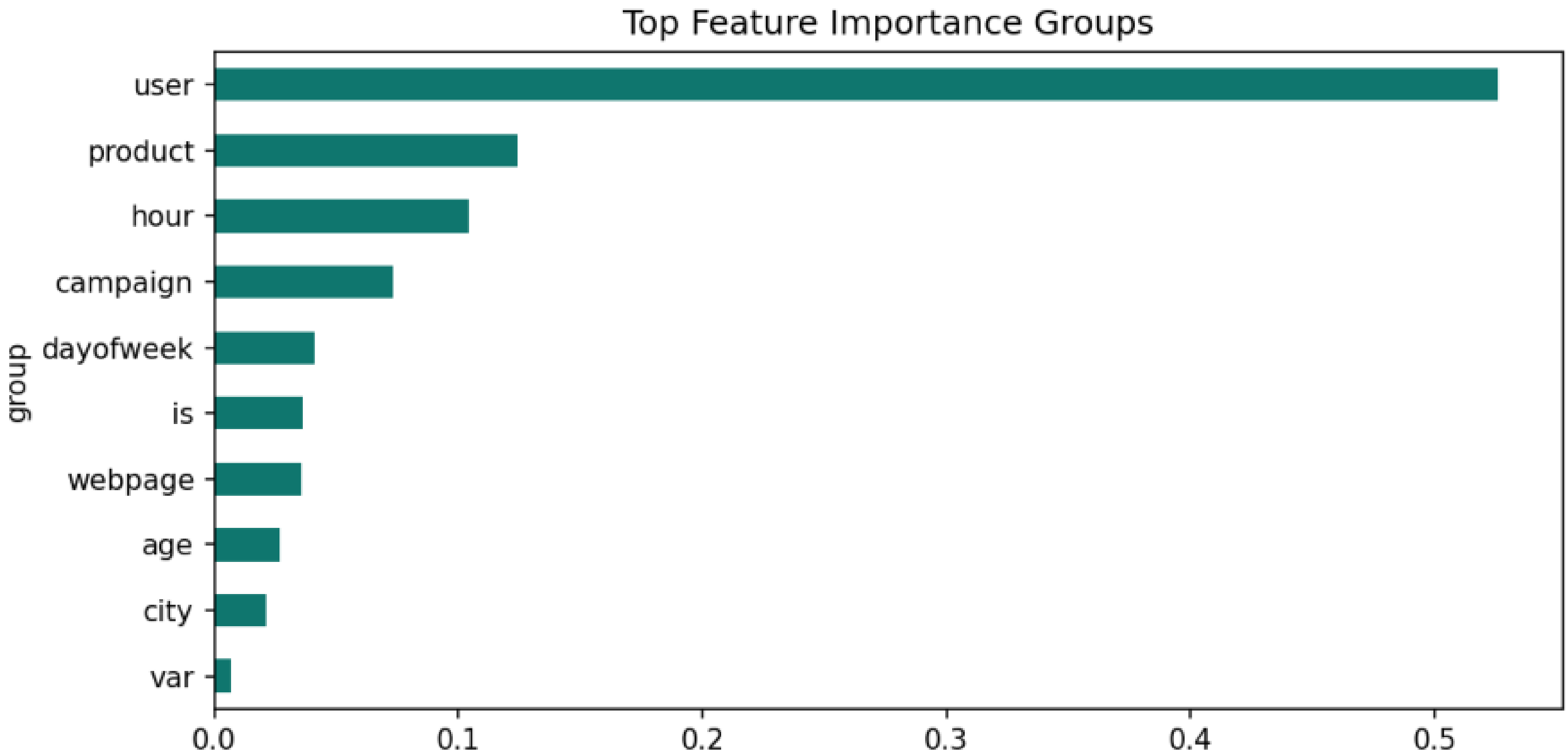
Model F1 Comparison



Product CTR



Feature Importance Groups



Product CTR and Inventory Forecasting Signal

product	clicks	impressions	ctr	expected_clicks_per_100k_impress
J	899	9698	0.0927	9,269.95
D	2949	41064	0.0718	7,181.47
H	7654	109574	0.0699	6,985.23
C	11306	163501	0.0691	6,914.94
E	1474	21452	0.0687	6,871.15
I	4079	63711	0.0640	6,402.35
A	953	15391	0.0619	6,191.93
B	1238	22479	0.0551	5,507.36
F	344	7007	0.0491	4,909.38
G	435	9414	0.0462	4,620.78

Use expected clicks per 100k impressions to reserve inventory and prioritize creative refresh cycles.

Highest Click-Propensity User Profiles

gender	age_level	city_development_index	mean	sum	count
Male	5.00	4.00	0.0866	255	2945
Female	5.00	4.00	0.0847	100	1181
Male	1.00	2.00	0.0764	605	7922
Female	5.00	2.00	0.0756	208	2752
Male	2.00	2.00	0.0741	3736	50409
Male	1.00	3.00	0.0732	607	8291
Male	1.00	4.00	0.0720	306	4252
Male	5.00	3.00	0.0711	349	4909
Male	2.00	3.00	0.0696	1927	27671
Male	5.00	2.00	0.0693	493	7112

Minimum 500 observations shown to reduce noisy micro-segment recommendations.

Weekend vs Weekday CTR

period	ctr	clicks	impressions
Weekday	0.0665	25540	384246
Weekend	0.0733	5791	79045

Weekend effect should be used as a monitored bid modifier, not a standalone rule.

Answers to Business Questions

? Weekend users: weekend CTR is 7.33% versus weekday CTR 6.65%.

? Highest CTR products: J, D, H. Lowest CTR products: B, F, G.

? Personalized interaction features help because they summarize affinity and context better than raw IDs.

? Main drivers include aggregate CTR features and contextual placement identifiers such as webpage, campaign, and product.

? SMOTE experiment recall: 0.426, F1: 0.133; useful offline, less attractive for low-latency serving.

? Aggregated product CTR forecasts expected clicks per inventory block and helps reserve impressions for top products.

? High-propensity profiles should receive bid uplifts only after volume, fairness, and privacy checks.

Strategic Recommendations

? Rank impressions by predicted click probability and tune the threshold by campaign budget and false-positive tolerance.

? Apply bid multipliers for high-CTR product, webpage, campaign, and qualified demographic-city segments.

? Reserve more inventory for products with high expected clicks per 100k impressions.

? A/B test personalized CTR features against a non-personalized baseline to measure incremental lift.

? Monitor ROC-AUC, PR-AUC, F1, recall, calibration, feature drift, and segment-level CTR weekly.

? Use weighted models for production refreshes; use SMOTE selectively for offline sensitivity analysis.